# Topic 2 — Animal Biology and Husbandry

**Sources baked in:** NCERT Class 9–11 (Animal Kingdom, Structural Organisation), Lucent General Science (Animal Husbandry, Breeds, Sericulture), 20th & 21st Livestock Census, ICAR breed compendiums, Ghatnachakra General Studies (Biology Subdivisions & Taxonomy 2025–1990), UPPCS Prelims PYQs (1990–2025), UKPCS, BPSC, RAS, and IAS Prelims **Weight:** ★★★ High — direct stems on egg-laying mammals, white blood / haemolymph, true vs false fish, snake fangs & venom, silk composition, cattle/buffalo breeds, diclofenac vulture ban and respiratory organs **Last verified:** September 2026

## Current Affairs

| Year | Event / Fact | Must-Score Fact |
|---|---|---|
| **2024–25** | **21st Livestock Census** field enumeration conducted from **October 2024 to February 2025** by Ministry of Fisheries, Animal Husbandry & Dairying. | Livestock census in India started in **1919** and is conducted every **5 years**; 20th census was in 2019 (India #1 in total livestock and buffalo population). |
| **2025** | Asiatic Lion Population Estimation (Gujarat Forest Department, May 2025) counted **891 lions**, up from **674 in 2020** (+32% growth). | Asiatic lion (*Panthera leo persica*) survives in the wild **only in the Greater Gir landscape** of Gujarat; listed as Endangered on IUCN Red List. |
| **2024** | **WPA 1972 Amendment (w.e.f. 2023) & Indian Flying Fox**: Schedule V (vermin list) omitted; Indian Flying Fox (*Pteropus giganteus*) moved to Schedule II; it is a frugivorous/nectar bat, NOT blood-sucking. | Section 62 allows Central Govt to declare Schedule II animals as vermin by notification; bats are vital pollinators and seed dispersers. |
| **2025** | **Peacock Tarantula (*Poecilotheria metallica*)** sighted in Pakkam Malai Reserve Forest (~370 km south of Eastern Ghats range); Critically Endangered, arboreal spider, NOT a crustacean. | Class Arachnida (8 legs, 2 body parts), insectivorous, threatened by habitat loss and pet trade. |
| **2006 / Still Current** | Government of India banned veterinary use of **Diclofenac** in **2006**; banned Ketoprofen and Aceclofenac in **2023**. | Non-Steroidal Anti-Inflammatory Drug (NSAID) given to cattle causes fatal visceral gout and acute renal failure in scavenging *Gyps* vultures. |
| **Current Status** | **National Milk Day** is celebrated on **26 November** to commemorate the birth anniversary of **Dr. Verghese Kurien** (Father of India's White Revolution). | India is the **world's largest milk producer** (~24% of global output), followed by the USA and Pakistan; leading Indian states are Uttar Pradesh and Rajasthan. |

## 2.1 Animal Nutrition & Digestive Strategies

- **Digestive System Adaptations Across Feeding Guilds**:
- **Herbivores**: Consume high-fibre plant matter rich in tough structural **cellulose**. Cellulose requires prolonged microbial fermentation, necessitating a **long and anatomically complex digestive tract**:
  - **Ruminants (Foregut Fermenters)**: Cattle, buffalo, sheep, goat, deer, giraffe.
  - **Non-ruminant Herbivores (Hindgut Fermenters)**: Horses, zebras, rabbits, elephants. Fermentation occurs in a massive, elongated **Caecum and Colon** located after the stomach and small intestine; less efficient than foregut fermentation (rabbits practice **coprophagy / caecotrophy** to re-ingest soft faeces for nutrient absorption).
- **Carnivores**: Consume nutrient-dense, easily digestible animal protein and fats; possess a **short, straight, simple digestive tract** with a reduced or rudimentary caecum (human caecum terminates in the vestigial vermiform appendix).
- **Omnivores**: Moderate intestinal length adapted for diverse dietary items (humans, bears, pigs, crows).
- **The Ruminant Compound Stomach Architecture**:
- Consists of four distinct chambers arranged sequentially:
  1. **Rumen (Paunch)**: The largest chamber (occupies ~80% of total stomach volume). Acts as a massive anaerobic fermentation chamber holding up to 150–200 litres; populated by billions of symbiotic anaerobic bacteria, protozoa, and fungi that break $\beta$-1,4-glycosidic bonds of cellulose into **volatile fatty acids** (acetate, propionate, butyrate).
  2. **Reticulum (Honeycomb)**: Inner lining shows hexagonal honeycomb-like cells; traps heavy foreign objects (nails, wire); coordinates regurgitation of coarse fibrous boli back into the mouth for re-chewing as **cud**.
  3. **Omasum (Manyplies / Bible)**: Contains numerous thin muscular leaves arranged like book pages; reabsorbs water, bicarbonate, and residual volatile fatty acids.
  4. **Abomasum (True Stomach)**: The only glandular, enzymatic compartment; secretes gastric juice containing **Hydrochloric Acid (HCl)** and **Pepsin** to digest microbial proteins and dietary matter.
- **The Camel Exception (Pseudoruminant Anatomy)**:
- Camels and llamas chew the cud and possess active ruminal microbial fermentation, but **completely lack the Omasum (third stomach compartment)**; their stomach is strictly 3-chambered.
- The camel's hump is composed entirely of **fat / adipose tissue**, not water; cellular oxidation of 1 gram of fat yields **1.07 grams of metabolic water**, but metabolic water generation requires oxygen intake which causes net respiratory water loss. Camels conserve water primarily through:
  - Highly concentrated, crystalline urine and extremely dry faeces.
  - Oval, nucleated RBCs capable of swelling up to 240% without haemolysing.
  - Wide tolerance of core body temperature fluctuations ($34^\circ C$ to $41^\circ C$), eliminating the need for sweating during daytime desert heat.

## 2.2 Respiration & Circulatory Dynamics in Animals

- **Evolution of Animal Respiratory Systems**:
- **Cutaneous Respiration**: Gas exchange directly across thin, vascular, moist body surface:
  - **Earthworms and Leeches**: Lack specialised lungs or gills; mucus and coelomic fluid keep the skin constantly moist; atmospheric $O_2$ dissolves in moisture and diffuses directly into sub-epidermal blood capillaries. If an earthworm's skin dries out, it suffocates and dies.
  - **Amphibians (Frogs)**: Frogs respire cutaneously during underwater submersion, winter sleep (**Hibernation**), and summer sleep (**Aestivation**).
- **Branchial Respiration (Gills)**:
  - Specialized vascularised external or internal outfoldings adapted for extracting dissolved oxygen from water:
  - Bony fish possess 4 pairs of gills housed in branchial chambers beneath a protective bony flap called the **Operculum**.
  - Cartilaginous fish (sharks, rays) possess 5–7 pairs of exposed gill slits lacking an operculum.
  - Counter-current exchange mechanism: Blood flows across gill lamellae in the opposite direction to water flow, ensuring maximum oxygen diffusion gradient (~80% extraction efficiency).
- **Tracheal Respiration (Direct Cellular Ventilation in Insects)**:
  - Air enters the body through lateral paired muscular valves called **Spiracles** (usually 10 pairs: 2 thoracic, 8 abdominal).
  - Spiracles lead into branched cuticular tubes called **Tracheae**, which further divide into fluid-filled microscopic **Tracheoles** penetrating directly into individual muscle cells.
  - **Crucial consequence**: Insect blood (haemolymph) **plays zero role in oxygen transport**; oxygen diffuses in the gas phase directly to cellular mitochondria.
- **Book Lungs & Book Gills**:
  - **Book Lungs**: Terrestrial arachnids (spiders, scorpions) possess internal chambers containing parallel thin vascular plates arranged like the leaves of a closed book.
  - **Book Gills**: Marine horseshoe crabs (***Limulus*** — an arthropod "living fossil") possess external respiratory plates arranged like pages.
- **Pulmonary Respiration (Lungs)**:
  - Found in terrestrial vertebrates: adult amphibians (buccopharyngeal + pulmonary), reptiles, birds, and mammals.
  - **Cetaceans (Whales and Dolphins)**: Aquatic placental mammals; lack gills; surface regularly to forcefully exhale humid warm air and inhale fresh air through modified dorsal nostrils called **Blowholes**; can hold breath underwater for up to 90 minutes due to massive muscle stores of **Myoglobin**.
- **Comparative Vertebrate Heart & Circulation**:

| Class | Heart Chambers | Circulation Type | Nature of Blood Flowing Through Heart |
|---|---|---|---|
| **Pisces (Fish)** | **2 Chambers** (1 Atrium + 1 Ventricle) | **Single Circulation** | Exclusively deoxygenated **Venous Blood** (pumps blood to gills for oxygenation, then directly to tissues) |
| **Amphibia (Frogs)** | **3 Chambers** (2 Atria + 1 Ventricle) | **Incomplete Double** | Oxygenated (left) and deoxygenated (right) blood mix inside the single ventricle |
| **Reptilia (Lizards, Snakes, Turtles)** | **3 Chambers** (Incompletely 4) | **Incomplete Double** | Ventricle has an incomplete interventricular septum; partial mixing occurs |
| **Crocodilia (Crocodiles, Alligators)** | **4 Chambers** (2 Atria + 2 Ventricles) | **Complete Double** | Complete anatomical separation of oxygenated and deoxygenated blood (**Exception among reptiles**) |
| **Aves (Birds)** | **4 Chambers** (2 Atria + 2 Ventricles) | **Complete Double** | Complete separation; right systemic aortic arch persists |
| **Mammalia (Mammals)** | **4 Chambers** (2 Atria + 2 Ventricles) | **Complete Double** | Complete separation; left systemic aortic arch persists |

## 2.3 The Invertebrate Spectrum: Phyla & Signatures

- **Phylum Porifera (Sponges / छिद्रधारी जन्तु)**:
- Most primitive multicellular animals; lack true tissues, organs, nerves, and muscles; asymmetrical or radially symmetrical.
- Body wall consists of outer **Pinacoderm** and inner **Choanoderm** separated by a gelatinous matrix (**Mesohyl**) containing amoebocytes and mineral **spicules** ($CaCO_3$ or silica) or organic **spongin fibres**.
- **Water Canal System**: Water enters through thousands of incurrent pores (**Ostia**) into a central cavity (**Spongocoel**) and exits through a terminal excurrent aperture (**Osculum**). Flagellated **Choanocytes (Collar cells)** generate water currents, trap microscopic food, and drive respiration and excretion.
- Examples: *Sycon* (*Scypha*), *Spongilla* (freshwater sponge), *Euspongia* (commercial bath sponge), *Euplectella* ("Venus flower basket", gifted as a wedding symbol of eternal union in Japan).
- **Phylum Cnidaria / Coelenterata (दंश कोशिका जन्तु)**:
- Aquatic, mostly marine, diploblastic, radially symmetrical animals with a central gastrovascular cavity (**Coelenteron**) opening via a single mouth/anus on a hypostome.
- Possess specialized stinging cells called **Cnidoblasts / Nematocysts** containing a coiled, venomous inverted thread (toxin: **Hypnotoxin**, a mix of proteins and phenols); function in offence, defence, and anchorage.
- Exhibit **Polymorphism**:
  - **Polyp**: Cylindrical, sessile, asexual form (e.g. *Hydra*, *Adamsia* sea anemone).
  - **Medusa**: Umbrella-shaped, free-swimming, sexual form with marginal tentacles (e.g. *Aurelia* jellyfish).

    - **Metagenesis (Alternation of Generation)**: In *Obelia*, polyps produce medusae asexually by budding, and medusae produce polyps sexually via fertilised eggs.

- Coral reefs are formed by massive colonies of stony corals (e.g. *Corallium rubrum* / red coral) that secrete massive skeletons of pure **Calcium Carbonate ($CaCO_3$)**.
- **Phylum Platyhelminthes (Flatworms / चपटे कृमि)**:
- Triploblastic, dorso-ventrally flattened, bilateral, acoelomate worms; organ-level organisation; incomplete gut (no anus).
- Excretion and osmoregulation performed by specialized ciliated **Flame Cells (Solenocytes)**.
- **Parasitic Adaptations**: Hooks, suckers, thick protective tegument, loss of digestive system, massive reproductive capacity:
    - ***Taenia solium*** (Pork Tapeworm): Lives in human small intestine; head (**scolex**) has 4 suckers and a rostellum with chitinous hooks; body consists of repeating segments called **proglottids**; causes Taeniasis and Cysticercosis.
    - ***Fasciola hepatica*** (Sheep Liver Fluke): Endoparasite in bile ducts of sheep and cattle; intermediate host is a freshwater snail (*Lymnaea*).
    - ***Planaria***: Free-living freshwater flatworm celebrated for its **immense, unmatched capacity of regeneration**.

- **Phylum Aschelminthes / Nematoda (Roundworms / गोलकृमि)**:
- Triploblastic, bilateral, unsegmented, vermiform worms with a cylindrical tapering body; possess a **Pseudocoelom** derived from the blastocoel.
- Complete digestive tract with a muscular, pumping pharynx.
- Show pronounced **Sexual Dimorphism**: Females are significantly longer and straight; males are shorter with a curved posterior tail bearing copulatory **pineal spicules**.
- Examples:
    - ***Ascaris lumbricoides***: Giant human intestinal roundworm; transmitted through water/food contaminated with embryonated eggs.
    - ***Wuchereria bancrofti***: Filarial worm; adult parasites obstruct lymph vessels of lower limbs and scrotum, causing massive chronic swelling called **Elephantiasis (Filariasis)**; transmitted by the bite of female ***Culex*** mosquitoes.
    - ***Ancylostoma duodenale*** (Hookworm) and ***Enterobius vermicularis*** (Pinworm).

- **Phylum Annelida (Segmented Worms / वलयी कृमि)**:
- Triploblastic, coelomate, bilateral animals characterized by internal and external **Metameric Segmentation**; body covered by a thin moist non-cellular cuticle.
- Locomotion via chitinous **Setae** (earthworms) or muscular lateral paddle-like **Parapodia** (marine *Nereis*).
- Excretion and osmoregulation via segmentally arranged coiled tubes called **Nephridia**.
- Closed circulatory system; blood contains haemoglobin (**erythrocruorin**) dissolved in plasma (lacks RBCs).
- Examples:
    - ***Pheretima posthuma*** (Indian Earthworm): Hermaphrodite; clitellum in segments 14–16; possesses 4–5 pairs of lateral hearts; improves soil fertility.

- ***Hirudinaria granulosa*** (Leech): Sanguivorous ectoparasite; secretes **Hirudin** anticoagulant.

- **Phylum Arthropoda (Jointed-Leg Animals / संधिपाद जन्तु)**:
- Largest animal phylum (over two-thirds of all named animal species, >1 million insects described); segmented body with jointed appendages and an external **Chitinous Cuticle** shed via **Ecdysis / Moulting**.
- Open circulatory system: haemolymph pumped into a **Haemocoel**.
- Sensory organs: Compound eyes with **Ommatidia** (mosaic vision), antennae, and statocysts.
- Excretion: **Malpighian tubules** (insects), **Green glands / Antennal glands** (prawns/crustaceans), **Coxal glands** (arachnids).
- **Insects (Hexapoda) vs Arachnids (Arachnida) — Fundamental Differentiation**:
  - **Class Insecta**: Body divided into 3 tagmata (**Head, Thorax, Abdomen**); **3 pairs of walking legs (6 legs)** attached to the thorax; usually 2 pairs of wings; 1 pair of antennae; tracheal respiration. Examples: Cockroach, Butterfly, Honeybee, Mosquito, Housefly, Cicadas, Froghoppers, Pond skaters, Termites.
  - **Class Arachnida**: Body divided into 2 regions (**Cephalothorax and Abdomen**); **4 pairs of walking legs (8 legs)**; **zero wings and zero antennae**; mouthparts consist of 1 pair of **chelicerae** (pincer-like fangs) and 1 pair of **pedipalps** (sensory/grasping). Examples: Spiders, Scorpions, Ticks, Mites.
  - **Ticks and Mites (Subclass Acari)**: Appendages jointed, body bilaterally symmetrical and triploblastic; individuals appear as an unsegmented single body mass (abdomen fused with cephalothorax).
  - **Peacock Tarantula / Gooty Tarantula (*Poecilotheria metallica*)**: Sighted in 2019 in Pakkam Malai Reserve Forest (expanding known range ~370 km south of Eastern Ghats); Critically Endangered, **arboreal (tree-dwelling), insectivorous spider, NOT a crustacean**.
  - **Spider Silk (Gossamer Silk)**: Protein fibre spun from abdominal spinnerets; miraculously combines high tensile strength (often compared to steel) with high elasticity.
  - **Sexual Cannibalism in Spiders**: Phenomenon where the female spider kills and consumes the male before, during, or after copulation (strong males occasionally escape to mate again).

- **Phylum Mollusca (Soft-Bodied Animals / कोमल शरीरी जन्तु)**:
- Second largest animal phylum; unsegmented soft body usually enclosed within a calcareous shell ( $CaCO_3$) secreted by a fleshy fold called the **Mantle (Pallium)**.
- Body divided into **Head**, ventral **Muscular Foot**, and dorsal **Visceral Hump** with a rasping tongue-like **Radula**.
- Key Representatives:
  - ***Pinctada vulgaris*** (Pearl Oyster, Class Bivalvia): Irritation from foreign particles induces mantle epithelium to secrete concentric sheets of nacre (aragonite $CaCO_3$ + conchiolin), forming genuine pearls.
  - ***Octopus*** (Devilfish, Class Cephalopoda): 8 flexible arms with suckers, bilaterally symmetrical, complex camera eyes, ink sac for defensive smoke screen; **no shell**.
  - **Squids**: **Colossal Squid (*Mesonychoteuthis hamiltoni*)** is the heaviest invertebrate; **Giant Squid (*Architeuthis dux*)** is the longest invertebrate (up to 18 metres).

- **Phylum Echinodermata (Spiny-Skinned Animals / शूलचर्मी जन्तु)**:
- Exclusively marine deuterostomes; adults show **pentamerous radial symmetry**, larvae are bilateral; possess a calcareous endoskeleton of ossicles/spines.
- **Water Vascular (Ambulacral) System**: Hydraulic network operating suction-cup **Tube Feet (Podia)** for locomotion and food capture.
- **Logic: All echinoderms are NOT viviparous — they are typically oviparous with external fertilisation and indirect development with free-swimming larvae.**
- Examples: *Asterias* (Starfish), *Echinus* (Sea urchin), *Holothuria* (Sea cucumber), *Ophiura* (Brittle star).
- **Feeding Strategies — Detritivores vs Filter Feeders**:
- **Detritivores**: Heterotrophs that feed on dead and decaying organic matter (detritus), aiding remineralisation: **Earthworms, Millipedes, Woodlice, Dung flies, Slugs**. (Seahorses and Jellyfish are secondary consumers/predators, NOT detritivores).
- **Filter Feeders (Suspension Feeders)**: Strain suspended particles and plankton from water via specialized filtering structures: **Oysters, Clams, Sponges, Baleen Whales, Flamingos**. Oysters draw water through gills and automatically filter nutrients, acting as vital aquatic ecosystem engineers.

## 2.4 Insects: Vectors, Products & Economic Roles

- **Insect Anatomy, Respiration & Haemolymph**:
- Head (compound eyes, antennae, feeding mouthparts), Thorax (3 segments, 3 pairs of legs, wings), Abdomen (viscera).
- **Insect Blood (Haemolymph)**: Colourless / white in cockroaches because it completely lacks respiratory pigments; carries nutrients and uric acid, but **zero oxygen**. Respiration is carried out entirely via lateral **Spiracles** leading into branching **Tracheae**.
- **Mosquito Biology — Gender Differences & Resting Posture**:
- **Diet & Mouthparts**:
  - **Female Mosquitoes**: Mouthparts adapted to **pierce and suck blood** from warm-blooded birds and mammals (ectoparasite); labium acts as a sheath, labrum forms the central feeding tube, and paired mandibles and maxillae form a sharp piercing stylet. Females require blood proteins to develop eggs.
  - **Male Mosquitoes**: Mouthparts lack piercing stylets; adapted strictly to **suck floral nectar and fruit juices**.
  - Females generally have noticeably **larger wings and body sizes** than males.
- **Resting Posture Differentiation**:
  - **Female *Anopheles***: Body rests at an **angle of ~45° to the substratum** with its abdomen pointing upward into the air.
  - **Female *Culex***: Body rests **parallel to the surface of the substratum**.
- **Bioluminescence & Enzyme Luciferase**:
- Glowworms (*Arachnocampa luminosa*) and fireflies emit blue-green light from abdominal light organs through the enzyme **Luciferase** oxidising luciferin in the presence of ATP and $Mg^{2+}$.
- **Beneficial vs Harmful / Destructive Insects**:

- **Beneficial Insects**: *Bombyx mori* (silk), *Kerria lacca* (lac resin), *Apis mellifera* (honey, beeswax, pollination).
- **Destructive Crop Pest**: ***Helicoverpa armigera*** (Cotton bollworm / American bollworm / Corn earworm); polyphagous pest attacking cotton, tomato, maize, and $> 100$ crops, causing severe economic damage.
- **Parasitoids vs Predators vs Detritivores**:
- **Parasitoids**: Lay eggs on or inside a host organism; developing larvae feed on the host's tissues and ultimately kill it: **Parasitic Wasps (Ichneumonids, Braconids), Parasitic Flies (Tachinids), and Carabid Ground Beetles**.
- **Generalist Predators**: Centipedes (carnivorous arthropods with poison claws).
- **Detritivores**: Termites (**White Ants**, infraorder Isoptera; eusocial insects that consume dead wood, cellulose, and leaf litter).
- **Ant Biology**:
- Ants inject **Formic acid ($HCOOH$)**, a simple carboxylic acid, when stinging/biting.
- Ants have the **largest brain in proportion to body size** in the entire animal kingdom.
- **Fungus Farming in Ants**: Leafcutter ants (tribe *Attini*) cut foliage to actively cultivate specialized fungal gardens in underground colonies as their exclusive food source.
- **Other Notable Insects**:
- **Cicadas (Superfamily Cicadoidea, Hemiptera)**: Sound-producing insects (>3,000 species) with stout bodies and clear wings.
- **Froghoppers (Superfamily Cercopoidea, Hemiptera)**: Powerful jumping insects; plant-sucking nymphs produce protective foam shelters (**spittlebugs**).
- **Pond Skaters / Water Striders (Family Gerridae, Hemiptera)**: Insects anatomically adapted to run/skate across the water surface film using surface tension, capturing insects that fall onto water.
- **Insects as Disease Vectors**:

| Vector Insect | Causative Pathogen | Disease Transmitted | Pathogen Nature |
|---|---|---|---|
| **Female *Anopheles* Mosquito** | *Plasmodium vivax / falciparum* | **Malaria** | Sporozoan Protozoan |
| **Female *Aedes aegypti* Mosquito** | Dengue virus (Flavivirus), Chikungunya, Zika | **Dengue**, **Chikungunya**, **Yellow Fever**, **Zika** | RNA Virus |
| **Female *Culex quinquefasciatus*** | *Wuchereria bancrofti* | **Filariasis (Elephantiasis)** | Nematode Roundworm |
| **Sandfly (*Phlebotomus argentipes*)** | *Leishmania donovani* | **Kala-azar (Visceral Leishmaniasis / Dum-Dum Fever)** | Flagellated Protozoan |
| **Tsetse Fly (*Glossina palpalis*)** | *Trypanosoma brucei* | **African Sleeping Sickness** | Flagellated Protozoan |
| **Housefly (*Musca domestica*)** | *Vibrio cholerae*, *Salmonella typhi*, *Entamoeba* | **Cholera**, **Typhoid**, **Amoebic Dysentery** | Mechanical Vector (Bacteria / Protozoa) |

| Vector Insect | Causative Pathogen | Disease Transmitted | Pathogen Nature |
|---|---|---|---|
| **Rat Flea (*Xenopsylla cheopis*)** | *Yersinia pestis* (Pasteurella) | **Bubonic Plague (Black Death)** | Bacterium |
| **Body Louse (*Pediculus humanus*)** | *Rickettsia prowazekii* | **Epidemic Typhus** | Bacterium (Rickettsia) |

> **Note:** UPPCS 2018 tested: Silk, honey, and lac are obtained from insects; **Pearl** is obtained from a mollusc (pearl oyster), **not** an insect.

## 2.5 Fish, Amphibians, Reptiles & Mammalian Exceptions

- **Class Pisces — True Fish vs False Fish**:
- **True Fish Characteristics**: Cold-blooded aquatic chordates, streamlined body with scales, breathe via gills, swim via paired/unpaired fins, 2-chambered venous heart.
- **Cartilaginous Fish (Chondrichthyes)**: Skeleton made purely of cartilage; **contain ZERO true bones** (Shark fish = 0 bones); ventral mouth; placoid scales; exposed gill slits without operculum; no air bladder: *Scoliodon* (Dogfish / Dog Shark), *Pristis* (Sawfish), *Torpedo* (Electric ray), *Trygon* (Stingray).
- **Bony Fish (Osteichthyes)**: Skeleton of bone; terminal mouth; cycloid scales; 4 pairs of gills covered by a bony **Operculum**; possess an **Air Bladder / Swim Bladder** regulating buoyancy:
  - *Hippocampus* (**Seahorse** — bony fish, vertical swimming, male carries fertilised eggs in brood pouch).
  - *Clarias* (**Catfish** — order Siluriformes, prominent whisker-like sensory barbels).
  - ***Tor putitora* (Golden Mahseer)**: Endangered cyprinid **fish** of rapid Himalayan rivers (frequently confused in PSCs as a bird; it is a true fish!).
  - *Exocoetus* (Flying fish), Pipefish, Paddlefish, Guitarfish, Globefish.
- **Master Fact — Animals Named "Fish" That Are NOT True Fish**:
  - **Starfish (*Asterias*)**: Phylum Echinodermata (no gills, fins, or scales).
  - **Jellyfish (*Aurelia*)**: Phylum Cnidaria / Coelenterata (diploblastic medusa).
  - **Silverfish (*Lepisma*)**: Phylum Arthropoda, Class Insecta (wingless nocturnal insect).
  - **Cuttlefish (*Sepia*)**: Phylum Mollusca, Class Cephalopoda (internal shell).
  - **Hagfish (*Myxine*)**: Jawless cyclostome chordate (lacks true jaws and paired fins).
  - **Devilfish (*Octopus*)**: Phylum Mollusca, Class Cephalopoda (8-armed mollusc).
  - **Crayfish**: Phylum Arthropoda, Class Crustacea (freshwater decapod crustacean).
- **Why Fishes Survive Under Frozen Lakes**:
  - In winter, ambient atmospheric temperature falls below $0°C$, freezing the surface layer into ice. Ice is a poor conductor of heat and floats on water.
  - Water has anomalous density, reaching maximum density at $4°C$. Denser water at $4°C$ sinks to the bottom, keeping bottom layers liquid ($> 0°C$), allowing fish and aquatic life to thrive beneath

the frozen surface.
    - When taken out of water, fish suffocate because delicate gill filaments stick together, collapsing the respiratory surface area and preventing oxygen diffusion.

- **Class Amphibia (Dual-Life Vertebrates)**:
- Ectothermic (cold-blooded); smooth, moist, glandular skin **completely devoid of scales**; 3-chambered heart (2 atria + 1 ventricle); tolerates partial mixing of oxygenated and deoxygenated blood.
- Breathe through moist skin (**cutaneous**) and lungs (**pulmonary**).
- **Metamorphosis**: Herbivorous aquatic tadpole larva undergoes abrupt structural changes controlled by **Thyroid Hormone**:
    - Gills disappear and lungs develop.
    - Tail is resorbed.
    - **Intestine dramatically shortens in length** (transitioning from a long herbivorous gut processing plant matter to a compact carnivorous gut digesting flesh/insects).

- **Class Reptilia (Creeping / Crawling Vertebrates)**:
- First truly terrestrial vertebrates; ectothermic; dry cornified skin covered by epidermal **scales or scutes**; lay cleidoic shelled eggs.
- Heart is **3-chambered** with an incomplete interventricular partition; **Crocodiles, Alligators, and Gharials possess a fully 4-chambered heart**.
- **Snakes & Snake Venom Dynamics**:
    - Limbless reptiles (loss of limbs explained by Lamarck's use-and-disuse principle).
    - **Poisonous Fangs**: Modified **maxillary teeth** (usually 1 pair) attached to venom ducts; hollow or grooved like hypodermic needles.
    - **Venom Glands**: **Homologous to salivary glands** of vertebrates; venom is chemically modified saliva containing 20+ protein toxins and enzymes.
    - **Venom Types**:
    - **Neurotoxic & Cardiotoxic**: Cobra (*Naja naja*) and Krait (*Bungarus caeruleus*); paralyzes respiratory muscles and attacks neuromuscular junctions.
    - **Vasculotoxic & Haemotoxic**: Russell's Viper (*Daboia russelii*) and Saw-scaled Viper (*Echis carinatus*); destroys vascular endothelium, lyses RBCs, and disrupts blood clotting.
    - **King Cobra (*Ophiophagus hannah*)**: Longest venomous snake ( $3.6\ \mathrm{m}$); the **ONLY snake in the world that builds an above-ground nest of leaves**; female deposits 20–50 eggs and guards the mound until hatching; diet consists almost exclusively of **other snakes** (rat snakes, kraits, small pythons).
    - **Python (Family Pythonidae)**: Giant non-venomous constrictor snake.
    - **Green Anaconda (*Eunectes murinus*)**: Viviparous snake (gives birth to live young rather than laying eggs).

- **Poisonous Organisms Across Classes**:
- Toxic species exist among **Butterflies** (African Giant Swallowtail, Monarch — sequester cardiac glycosides from larval host plants), **Fish** (Puffer fish contains lethal tetrodotoxin; Reef stonefish is most venomous fish), and **Frogs** (Poison dart frogs store batrachotoxin alkaloids in skin).

- **Class Aves (Birds / Feathered Bipeds)**:
- Warm-blooded (homeothermic); 4-chambered heart; pneumatic bones; toothless beak; right aortic arch.
- **Kiwi (*Apteryx*)**: Small flightless bird endemic to New Zealand.
- **White Leghorn**: World's leading commercial egg-laying breed of fowl (chicken).
- Indian Nightjar, Spoonbill, White Ibis belong to Aves.
- **Class Mammalia & Mammalian Exceptions**:
- Warm-blooded, mammary glands, body hair, diaphragm, 7 cervical vertebrae, 4-chambered heart.
- **Blue Whale (*Balaenoptera musculus*)**: Largest animal ever known ($33\ \mathrm{m}$, $> 180\ \mathrm{tonnes}$); lifespan 80–90 years; aquatic mammal.
- **Sperm Whale (*Physeter macrocephalus*)**: Largest of the **toothed whales** ($55\ \mathrm{ft}$, $35–45\ \mathrm{tonnes}$); carnivorous predator; possesses the **largest absolute brain in the animal kingdom (~7 kg)**.
- **Cetaceans (Whales, Dolphins, Porpoises)**: Breathe air via dorsal nostrils (**Blowholes**); warm-blooded; nurse calves with milk. **Ganges River Dolphin (*Platanista gangetica*)** is India's National Aquatic Animal.
- **Semiaquatic Mammals**: Seal, Sea Lion, Walrus (order Carnivora, pinnipeds). **Sea cow (Dugong)** is a giant herbivorous aquatic mammal (order Sirenia).
- **Bats (Order Chiroptera)**: **Only mammals naturally capable of true sustained powered flight** via skin membrane wings (**patagium**). Smallest mammal: **Kitti's hog-nosed bat (bumblebee bat)** ($29–34\ \mathrm{mm}$); largest bat: Flying fox (wingspan $1.7\ \mathrm{m}$). **Indian Flying Fox (*Pteropus giganteus*)** is frugivorous (eats fruits and nectar), NOT blood-sucking.
- **Egg-Laying Mammals (Prototheria / Monotremes)**: Lay soft leathery eggs; have cloaca and mammary glands without nipples: **Duck-billed Platypus** (*Ornithorhynchus anatinus*) and **Spiny Anteater / Echidna** (*Tachyglossus*).
- **Pouched Mammals (Metatheria / Marsupials)**: Give birth to immature altricial young nurtured in abdominal pouch (**Marsupium**): Kangaroo, Koala, Opossum, Wombat, Tasmanian devil, Wallaby, Bandicoot. **Marsupials are NOT naturally found in India**; they occupy diverse habitats (deserts, forests, savannas), NOT exclusively montane grasslands.
- **Desert Adaptations**:
  - **Kangaroo Rat (*Dipodomys*)**: Can live its entire life without drinking water; produces the **most concentrated, almost solid urine** among mammals; lives on metabolic water from oxidized seeds.
  - **Camel**: Stores water in its **intestine** (drinks $> 100\ \mathrm{litres}$ at once) and stores energy as fat in its hump; broad padded 2-toed feet; pseudoruminant (lacks omasum).
- **Absence of Gallbladder**: Animals lacking a gallbladder include: all members of the **deer family (except musk deer)**, **equines (horses, zebras)**, camels, giraffes, elephants, rhinoceroses, whales, doves, and rats.
- **Hooves vs Claws**: Horses, cattle, deer have hooves; **Jackal (*Canis aureus*)** belongs to Canidae and has **padded feet with claws, NO hooves**.
- **Nilgai (*Boselaphus tragocamelus*)**: Largest Asian **antelope** (family Bovidae, hollow permanent horns); NOT a cow.
- **Big Cat Wildlife Facts**:

    - Scientific names: Leopard (***Panthera pardus***), Lion (*Panthera leo*), Tiger (*Panthera tigris*), Snow Leopard (*Panthera uncia*), Jaguar (*Panthera onca*).
    - Lions have **no specific breeding season** (breed year-round in captivity, every 2 years in wild).
    - **Cheetahs do NOT roar** (they growl, chirp, purr, and meow).
    - Male and female leopards proclaim territory via **scent-marking and urine spraying**.
- **Self-Defense Mechanisms**:
    - **Hedgehog**: Spiny mammal that rolls into a tight ball with muscular control pointing quills outward.
    - **Pangolin (Scaly Anteater)**: Rolls into an armoured ball with sharp overlapping keratin scales protecting face.
    - **Marmots**: Large herbivorous ground squirrels; **do NOT roll up**; escape predators by diving into underground burrows.
- **Indian Squirrels**: Arboreal; **build nests (dreys) in trees** (not burrows in ground); bury/hoard nuts and seeds in soil; strictly **omnivorous** (eat nuts, fruits, insects, bird eggs).
- **Primate Evolution & Behaviour**:
    - **Apes vs Monkeys**: Apes (Superfamily Hominoidea) are **tailless**: Gibbons (lesser apes), Chimpanzees, Gorillas, Orangutans, Humans (great apes). **Gibbon** is the smallest, gentlest man-like ape. **Langurs** have long tails and are Old World monkeys (Colobinae), NOT apes.
    - **Tool Use**: **Orangutan** uses sticks to scrape insects from tree holes, poke logs for honey, and open irritating *Neesia* fruits.
    - **Language**: Chimpanzee, Parrot, and Dolphin have been trained to use language symbols with success; **dogs cannot learn language**.
    - **Human Distinctiveness**: Humans are the only animals evincing a conscious desire to learn, philosophise, and express complex symbolic thought.
    - **Cranial Capacity**: Modern human average cranial cavity is **~1,350 cc** (range 1,000–2,000 cc); fossil **Cro-Magnon man had ~1,600 cc**. **Ant has the largest brain in proportion to its body weight**.
    - **Giant Panda**: Member of the bear family (**Ursidae**).
    - **Dingo (*Canis lupus dingo*)**: Australian wild dog that **cannot bark**; howls like a wolf.
    - **Sloth bear**: Common street performing bear in India.
    - **Otter**: Semiaquatic mustelid (subfamily Lutrinae); aggressive aquatic predator.
- **Summary of Mammalian Sub-Classes**:
    1. **Prototheria (Monotremes / Egg-Laying)**: Platypus (*Ornithorhynchus*) and Spiny Anteater (*Echidna*).
    2. **Metatheria (Marsupials / Pouched)**: Kangaroo, Koala, Opossum, Wombat (abdominal marsupium; not naturally found in India).
    3. **Eutheria (Placental)**: Whales (*Balaenoptera*), Bats (*Pteropus*), Elephants, Primates (nourished in utero via chorio-allantoic placenta).

---

## 2.6 Silk, Sericulture & Animal Products

- **Biology of Silk & The Silkworm Lifecycle**:
- Silk is a continuous animal protein filament spun by the mature caterpillar larva of silk moths during cocoon construction:
- **Life Cycle Stages**:
  1. **Egg**: Female moth lays 300–500 eggs.
  2. **Larva (Caterpillar)**: Voracious feeder on host leaves; undergoes 4 moults through 5 instars over 25–30 days.
  3. **Pupa / Chrysalis**: Fully grown 5th instar larva stops feeding, anchors itself, and rotates its head in a figure-8 motion, extruding a liquid secretion from paired labial silk glands through a spinneret. The liquid solidifies upon contact with air into a continuous silk filament ( $1000–1500\ \text{metres}$ long), forming an oval protective shell called the **Cocoon**.
  4. **Adult Moth**: In nature, the pupa metamorphoses into a moth that secretes an alkaline fluid to dissolve cocoon threads and emerges, rendering the broken thread unusable for reeling.
- **Stifling & Reeling**: In commercial silk production, live pupae inside cocoons are killed by hot air, steam, or baking before emergence (**Stifling**); cocoons are then soaked in hot water to soften sericin gum, and continuous fibres from multiple cocoons are wound together onto reels (**Reeling**).
- **Chemical Nature of Natural Silk**:
- **Fibroin (75–80%)**: Insoluble crystalline core structural protein composed predominantly of the amino acids **Glycine, Alanine, and Serine** in anti-parallel $\beta$-pleated sheets; gives silk its exceptional tensile strength and softness.
- **Sericin (20–25%)**: Amorphous, water-soluble globular sticky gelatinous protein that coats and cements the two fibroin filaments together.
- **Logic:** Natural silk is a pure protein and therefore **always contains Nitrogen** (~17%); cotton contains zero nitrogen because it is pure plant carbohydrate (cellulose). Spider dragline silk is also an exceptionally strong protein.
- **The Four Commercial Silks of India**:
- **Mulberry Silk**: Reared on cultivated White Mulberry (*Morus alba*); softest, finest, white or creamy yellow; accounts for **~70% of total Indian silk production**; **Karnataka is #1** (Mysuru, Mandya, Ramanagara), followed by Andhra Pradesh and West Bengal. Central Silk Board headquarters is located at **Bengaluru**.
- **Tasar Silk (Tussar)**: Wild forest vanya silk obtained from ***Antheraea mylitta***; caterpillars feed on leaves of Arjun (*Terminalia arjuna*) and Asan (*Terminalia tomentosa*); coarse, copper-brown, porous; **Jharkhand is #1** (encapsulating Santhal Parganas and Chaibasa), followed by Chhattisgarh and Odisha.
- **Muga Silk**: Exclusive to the state of **Assam**; produced by ***Antheraea assamensis*** feeding on aromatic leaves of Som (*Machilus bombycina*) and Soalu (*Litsaea polyantha*); famous for its **natural brilliant golden-yellow luster**, exceptional durability, and glossy texture (improves with washing); held by Assam as a protected Geographical Indication (GI Tag).
- **Eri Silk (Endi)**: Spun silk produced by ***Samia ricini*** feeding on Castor leaves (*Ricinus communis*); white or brick-red; thermal properties comparable to wool; called **"Ahimsa Silk"** because the cocoon

is open-ended, allowing the adult moth to emerge naturally without boiling the live pupa. Primary states: Assam and Meghalaya.

---

## 2.7 Animal Husbandry & Elite Indian Breeds

- **Livestock Dynamics in India**:
- Livestock Census is conducted every **5 years** by the Department of Animal Husbandry and Dairying; the first census was conducted in **1919**.
- According to the **20th Livestock Census (2019)**:
  - Total Livestock Population = **535.78 million** (+4.6% over 2012).
  - Total Bovine (Cattle + Buffalo + Mithun + Yak) = **302.79 million**.
  - India ranks **#1 in the world in total livestock population, buffalo population, and milk production**.
  - Top livestock states: Uttar Pradesh > Rajasthan > Madhya Pradesh.
- **Artificial Insemination**: Collection, evaluation, dilution, and storage of superior bull semen; stored indefinitely in liquid nitrogen tanks at $-196^{\circ}C$ in cryoprotectant media containing egg yolk and glycerol.
- **Milch, Draught & Dual Cattle Breeds (*Bos indicus*)**:
- **Milch Breeds (High Dairy Yield)**:
  - **Sahiwal**: Montgomery district (Punjab/Pakistan); heavy body, loose skin (termed *Lola*), reddish-dun coat; yields **2000–3000 litres per lactation**; premier dairy breed.
  - **Gir**: Gir forest of Kathiawar (Gujarat); prominent rounded convex forehead, long pendulous drooping ears folded like an inverted leaf; highly docile.
  - **Red Sindhi**: Karachi and Hyderabad (Sindh); deep red colour; high heat tolerance and tick resistance.
  - **Deoni**: Latur and Marathwada (Maharashtra); black-and-white spotted coat; resembles Gir.
- **Draught Breeds (Heavy Agricultural & Transport Work)**:
  - **Amritmahal**: Hassan and Mysore (Karnataka); slender, active, greyish coat, long sharp horns; legendary endurance, historically used for army haulage by Tipu Sultan.
  - **Hallikar**: Karnataka; compact body, long dark curved horns; ancestral progenitor of Amritmahal.
  - **Nagori**: Nagaur (Rajasthan); agile, light-footed trotting bullocks.
  - **Malvi**: Malwa plateau (Madhya Pradesh); massive draught power.
  - **Kangayam**: Coimbatore (Tamil Nadu); short, stout draught animal.
- **Dual-Purpose Breeds (Work Bullocks + Fair Milk Yield)**:
  - **Hariana**: Rohtak, Hisar, Jind, Gurgaon (Haryana); cows yield 1000–1500 litres; bullocks are excellent plow and cart animals.
  - **Tharparkar (White Sindhi)**: Thar desert (Jodhpur/Jaisalmer, Rajasthan); medium frame, lyre-shaped horns; withstands extreme arid heat and scarcity of fodder.
  - **Kankrej**: Rann of Kutch (Gujarat) and neighbouring Rajasthan; **heaviest and largest indigenous cattle breed**; distinctive majestic gait termed **"Sawai Chal"**; lyre-shaped horns; basis of the

famous American Brahman cattle line.
    - **Ongole**: Prakasam and Guntur (Andhra Pradesh); large muscular humped bull; prized internationally for heat and tropical tick resistance.

- **Exotic Dairy Breeds**:
    - **Holstein-Friesian**: Netherlands; black and white patches; **highest volume milk yielder in the world** ($6000–8000$ litres per lactation); milk fat is moderate (~3.4–3.6%).
    - **Jersey**: Island of Jersey (UK); compact, light fawn colour; **highest milk fat percentage among dairy cattle (~5.0–5.5%)**; highly heat tolerant among exotics.
- **Elite Buffalo Breeds (*Bubalus bubalis*)**:
- **Murrah**: Rohtak, Hisar, Jind (Haryana) and Punjab; jet-black coat with white switch on tail; tightly curled spiral horns resembling a ram's; yields 2000–3000 litres of rich milk; universally hailed as **"Black Gold"**.
- **Bhadawari**: Ravines of Yamuna and Chambal (Agra, Etawah in UP; Bhind, Morena in MP); medium-sized, copper-coloured coat; wedge-shaped body; **highest fat content of all buffaloes (up to 12.5–13%)**; ideal for ghee production.
- **Jaffarabadi**: Gir forest, Saurashtra (Gujarat); massive heavy body, forehead shows a prominent frontal shield, horns droop down and curl outward; yields high milk volumes.
- **Surti**: Anand, Kheda, Vadodara (Gujarat); medium-sized, sickle-shaped horns with white markings.
- **Mehsana**: Mehsana, Banaskantha (Gujarat); synthesized stabilized cross of Murrah bull × Surti female; early maturing and regular breeder.
- **Nili-Ravi**: Ferozepur (Punjab) and Pakistan; distinctive walled / glass-white eyes (**"Panch Kalyani"**: white switch, white forehead, white muzzle, white legs, white eyes).
- **Elite Goat & Sheep Breeds**:
- **Jamunapari Goat**: Etawah (Chambal ravines, UP); largest Indian dairy goat breed; white coat with tan patches, long drooping ears ($25–30$ cm), convex Roman nose giving a parrot-mouth appearance; dual-purpose (milk and meat).
- **Barbari Goat**: Urban and peri-urban plains of UP, Haryana, and Delhi; small, compact, short erect ears, highly prolific (twins/triplets common); ideal for stall-fed intensive dairying ("City Goat").
- **Black Bengal Goat**: West Bengal, Bihar, Odisha; small, black coat; prized for high prolificacy (kidding twice a year) and world-class tender chevon meat and fine leather.
- **Changthangi Goat (Pashmina Goat)**: High-altitude arid plateaus of Changthang (Ladakh, $> 4000$ metres altitude); winter undercoat yields ultra-fine, warm, luxurious **Pashmina Cashmere** fibre ($12–14$ microns diameter).
- **Merino Sheep**: Native to Spain, commercialised in Australia; finest, softest apparel wool in the world with high crimp.
- **Marwari Sheep**: Rajasthan desert; black face, coarse carpet wool, hardy walker.
- **Nellore Sheep**: Andhra Pradesh; tallest sheep breed in India; hairy coat, raised exclusively for meat (mutton).
- **Poultry Science & Breeds**:
- **Kadaknath (Kalamasi)**: Indigenous chicken of Jhabua and Dhar districts of Madhya Pradesh; skin, beak, wattles, tongue, meat, bones, and visceral peritoneum are all **jet black** due to intense

deposition of the pigment **Melanin** (fibromelanosis); meat is rich in protein (~25%), iron, and essential amino acids, with very low fat and cholesterol; awarded a GI Tag.

- **Aseel**: Indigenous to Andhra Pradesh and Telangana; massive muscular frame, erect posture, strong beak; celebrated for fighting stamina and ferocious courage; raised for meat and gamecock sport.
- **White Leghorn**: Small, light-framed Mediterranean exotic bird; premier commercial egg-laying machine (layer), producing 280–300 white eggs per year.

## 2.8 Wildlife Conservation, Vultures & Diclofenac Crisis

- **The Indian Vulture Crisis & Veterinary Diclofenac**:
- Between the early 1990s and 2007, populations of India's native *Gyps* vultures (White-rumped vulture *Gyps bengalensis*, Indian vulture *Gyps indicus*, and Slender-billed vulture *Gyps tenuirostris*) plummeted by **over 99%** (from tens of millions to a few thousand), representing the fastest catastrophic population crash of any bird species in recorded history.
- **Causative Agent**: Non-Steroidal Anti-Inflammatory Drug (NSAID) **Diclofenac**, widely administered to cattle and buffaloes as a routine analgesic and painkiller for mastitis and arthritis.
- **Pathology**: Scavenging vultures feeding on the carcasses of treated livestock within 48–72 hours of administration ingested toxic doses; vultures lack the enzyme to metabolise diclofenac, leading to acute kidney failure, catastrophic hyperuricaemia, and extensive chalky white precipitation of uric acid crystals on internal organs (**Visceral Gout**), causing death within days.
- **Regulatory Ban**: Government of India formally **banned veterinary manufacture, sale, and use of Diclofenac in 2006**. In **2023**, the central government further banned veterinary use of **Ketoprofen and Aceclofenac** (which metabolises into diclofenac in cattle). **Meloxicam and Tolfenamic acid** are safe vulture-friendly alternatives.
- Conservation actions: Vulture Conservation and Breeding Centres (VCBC) established at Pinjore (Haryana), Buxa (WB), and Rani (Assam) by BNHS.
- **Asiatic Lion Conservation in Gir**:
- The Asiatic Lion (*Panthera leo persica*) is genetically distinct from African lions (smaller mane in males, longitudinal fold of skin along the belly).
- Survives in the wild strictly in the **Gir National Park and Wildlife Sanctuary** and surrounding agro-pastoral landscapes of Saurashtra (Junagadh, Amreli, Bhavnagar, Gir Somnath in Gujarat).
- Population increased from a critical bottleneck of ~20 individuals in the early 20th century to **891 individuals in May 2025** (up from 674 in 2020); Project Lion is active for habitat expansion (Barda Wildlife Sanctuary proposed as second home).
- **IUCN Red List Categories**:
- **Extinct (EX)**: Beyond reasonable doubt that the last individual has died (e.g. Dodo, Passenger Pigeon, Tasmanian Tiger / Thylacine, Indian Cheetah declared extinct in India in 1952).
- **Extinct in the Wild (EW)**: Survives only in cultivation, captivity, or as a naturalized population outside historic range.
- **Critically Endangered (CR)**: Faces an extremely high risk of extinction in the wild (e.g. Great Indian Bustard, Gharial, White-rumped Vulture, Hangul / Kashmir Stag).

- **Endangered (EN)**: Very high risk of wild extinction (e.g. Asiatic Lion, Royal Bengal Tiger, Asian Elephant, Gangetic Dolphin, Snow Leopard, Red Panda).
- **Vulnerable (VU)**: High risk of endangerment in the wild (e.g. One-horned Rhinoceros, Nilgiri Tahr).

## Bilingual Terminology -- Animal Biology and Husbandry

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Vertebrate** (कशेरुकी) | कशेरुकी | Animals with a backbone/vertebral column |
| **Invertebrate** (अकशेरुकी) | अकशेरुकी | Animals without a backbone (insects, worms, molluscs) |
| **Mammal** (स्तनधारी) | स्तनधारी | Warm-blooded; nurse young with milk; has hair/fur |
| **Amphibian** (उभयचर) | उभयचर | Cold-blooded; live on land and water; moist skin; e.g., frog |
| **Reptile** (सरीसृप) | सरीसृप | Cold-blooded; dry scaly skin; lay eggs on land; e.g., lizard |
| **Poikilotherm** (असमतापी) | असमतापी / शीतरक्तीय | Variable body temperature (cold-blooded); fish, amphibians, reptiles |
| **Homeotherm** (समतापी) | समतापी / उष्णरक्तीय | Constant body temperature (warm-blooded); birds and mammals |
| **Ruminant** (जुगाली करने वाला) | रोमन्थी | Herbivore with 4-chambered stomach; e.g., cow, buffalo, goat |
| **Husbandry** (पशुपालन) | पशुपालन | Rearing and breeding of animals for agricultural/commercial purpose |
| **Apiculture** (मधुमक्खी पालन) | मधुमक्खी पालन | Rearing honey bees for honey and wax |
| **Sericulture** (रेशम उत्पादन) | रेशम उत्पादन / सेरीकल्चर | Rearing silkworms for silk; Bombyx mori is common silk moth |
| **Pisciculture** (मत्स्य पालन) | मत्स्य पालन | Cultivation of fish in tanks/ponds |
| **Poultry** (कुक्कुट पालन) | कुक्कुट पालन | Rearing birds (hen, duck, turkey) for eggs and meat |
| **Lac Culture** (लाख उत्पादन) | लाख पालन | Rearing Kerria lacca insect for lac/shellac resin |
| **Pisces** (मत्स्य) | मत्स्य | Fish class; ectothermic; breathe through gills; 2-chambered heart |
| **Aves** (पक्षी) | पक्षी | Birds; warm-blooded; pneumatic bones; 4-chambered heart; feathers |
| **Exoskeleton** (बाह्य कंकाल) | बाह्यकंकाल | Hard outer covering of insects, crustaceans (chitin in arthropods) |
| **Metamorphosis** (कायांतरण) | कायांतरण | Transformation from larva to adult (egg-larva-pupa-adult) |
| **Parthenogenesis** (अनिषेकजनन) | अनिषेकजनन | Development from unfertilised egg; drone bees develop this way |

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Zoonosis** (जूनोसिस) | जूनोटिक रोग | Disease transmissible from animals to humans; e.g., rabies, plague |

## One-Liner Revision -- Animal Biology and Husbandry

| # | Fast Fact |
|---|---|
| 1 | Largest animal = Blue whale (Balaenoptera musculus) |
| 2 | Fastest land animal = Cheetah (~120 km/h) |
| 3 | Largest land animal = African elephant |
| 4 | Tallest animal = Giraffe |
| 5 | Only mammal to fly = Bat (Order Chiroptera) |
| 6 | Mammals that lay eggs (Monotremes) = Duck-billed platypus + Echidna |
| 7 | 4-chambered heart = Mammals, Birds, Crocodile |
| 8 | 3-chambered heart = Most amphibians and reptiles |
| 9 | 2-chambered heart = Fish (Pisces) |
| 10 | Ruminant stomach chambers: Rumen, Reticulum, Omasum, Abomasum |
| 11 | Largest egg (non-extinct) = Ostrich egg |
| 12 | Silk from = **Bombyx mori** (mulberry silkworm) |
| 13 | Lac produced by = **Kerria lacca** (lac insect) |
| 14 | Honey bee: Queen = fertilised egg; Drone = **unfertilised egg** (parthenogenesis) |
| 15 | Largest bee colony queen = **Queen bee** (lives 3-5 years); Workers live 6 weeks |
| 16 | Pearl produced by = Oyster (Mollusca) -- NOT an insect product |
| 17 | Coral = formed by **polyps** (phylum Cnidaria); calcium carbonate exoskeleton |
| 18 | Sponges = phylum Porifera; simplest multicellular animals; no tissues |
| 19 | Earthworm = hermaphrodite; no legs; locomotion by setae (bristles) |
| 20 | Starfish (Sea star) = NOT a true fish; phylum Echinodermata |
| 21 | Whale = mammal (not fish); breathes air; warm-blooded |
| 22 | Bat = mammal; only true flying mammal; uses echolocation (ultrasound) |
| 23 | Snake has no legs; moves using ribs + ventral scales; no eyelids |
| 24 | Chameleon = changes colour for temperature regulation + communication |

| # | Fast Fact |
|---|---|
| 25 | India's only ape = **Hoolock Gibbon** (NE India; Assam) |
| 26 | National Animal of India = **Bengal Tiger** (Panthera tigris tigris) |
| 27 | Asiatic Lion = found ONLY in **Gir Forest, Gujarat** |
| 28 | One-horned Rhinoceros = **Kaziranga NP, Assam** |
| 29 | Project Tiger launched = **1973** (first reserve = Jim Corbett NP) |
| 30 | Amphibians = first vertebrates to live on land; "biological thermometers" of ecosystem |